Monetary Policy Transmission and Building Land Scarcity: A Panel Local Projections Analysis of Regional Austrian Housing Markets

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Abstract

Acknowledgements

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1. Introduction

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4. Data and Descriptive Evidence

The empirical analysis rests on a balanced quarterly panel of Austrian districts covering the period Q1 2017 to Q4 2025. The political district is the spatial unit. The panel

therefore comprises 116 districts observed over 36 quarters, which yields 4 176 district-quarter observations. Vienna is retained at the level of its individual municipal districts, so that the Viennese housing market enters the analysis at the same granularity as the remaining political districts rather than as a single aggregate. The following subsections describe the outcome variable, the monetary policy shock, the supply-constraint measure, and the control variables in turn. Further, the subsections document descriptive evidence that motivates the following empirical analysis, before the closing subsection documents how these components are combined into the estimation panel along with preliminary diagnostics of the data.

4.1. Real Estate Price Data

Residential real estate prices are provided by "DataScience Service GmbH" (2026), which compiles district-level price measures from transaction data. The data distinguish three property categories, average prices for single-family houses (EFH), average prices for condominium apartments (ETW), and building land prices, each expressed in € per m². These data are based on land-register transaction records ("Grundbuchauszüge"). Building land is not treated as a housing outcome and is therefore excluded from the price series used in estimation. Over the full sample the price data are complete, with an observation for every district in every quarter and no missing values in any of the three categories.

The primary outcome is a district-level composite price constructed as a fixed-weight average of the house and apartment series. For district i in quarter t, the composite price level is defined as

$$p_{i,t}^{\text{comp}} = w_i^{\text{house}} \, \text{EFH}_{i,t} + w_i^{\text{apt}} \, \text{ETW}_{i,t}, \tag{4.1}$$

where $\text{EFH}_{i,t}$ and $\text{ETW}_{i,t}$ are the average house and apartment prices in € per m², and the weights w_i^{house} and w_i^{apt} are held constant over the sample and are derived from the district housing stock, as described in the next paragraph. The underlying price data are in nominal terms. No district-level deflator is available at quarterly frequency, but this poses no difficulty for the analysis, since the quarter fixed effects introduced in Section 5.5 absorb national inflation along with every other development common to all districts in

a quarter. The composite is formed at the level of prices and with fixed weights, so that cross-district property composition is held constant and only the underlying prices vary over time. This construction motivates the property-specific alternative specification described in Section 5.7, where the house and apartment series are examined separately, and it provides the price series that the outcome transformation in Section 5.3 then converts into the dependent variable.

The composite weights are based on the building stock recorded in the 2011 Statistik Austria building and housing census, which reports the number of dwellings by the number of dwellings within the building on the district-level (Statistik Austria, 2026c). Let D_i^1, D_i^2, $D_i^{3\text{–}5}$, $D_i^{6\text{–}10}$, $D_i^{11\text{–}20}$, and D_i^{21+} denote the 2011 dwelling counts in district i across the six building-size classes. Dwellings in single-dwelling and two-dwelling buildings are assigned to the house category, the latter to capture semi-detached and duplex dwellings that behave like the owner-occupied house market, while dwellings in buildings with three or more dwellings are assigned to the apartment category. The house weight is then the house-category share of the district dwelling stock,

$$w_i^{\text{house}} = \frac{D_i^1 + D_i^2}{D_i^1 + D_i^2 + D_i^{3\text{–}5} + D_i^{6\text{–}10} + D_i^{11\text{–}20} + D_i^{21+}}, \qquad w_i^{\text{apt}} = 1 - w_i^{\text{house}}. \tag{4.2}$$

The weights are fixed at their 2011 values so that they predate the estimation window and remain exogenous to price developments during the sample, following the same logic that governs the supply-constraint measure below.

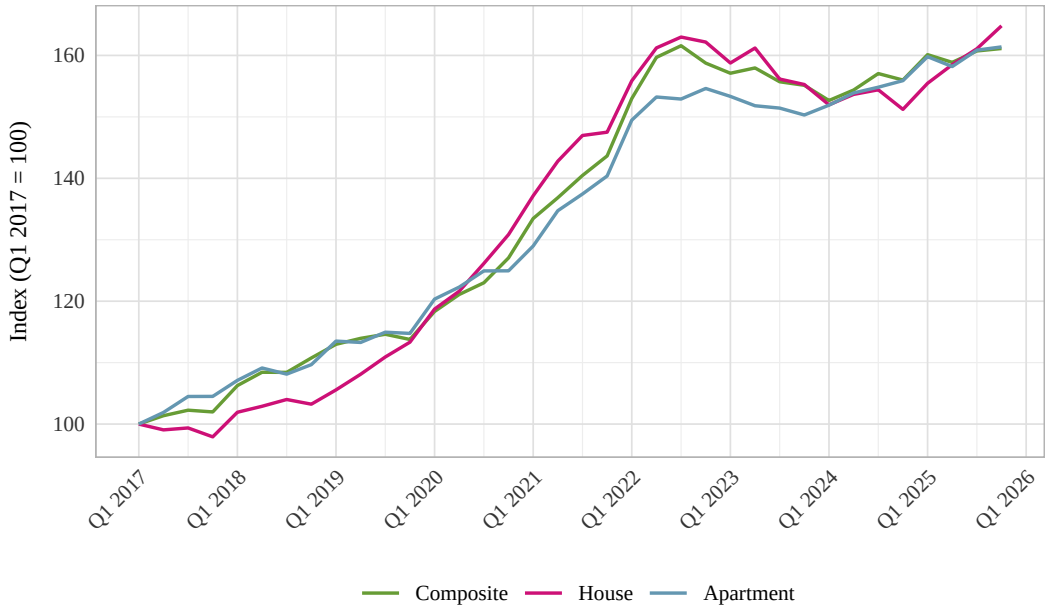


Figure 1: District price indices by property type
Unweighted district medians, Q1 2017 = 100.

Across the sample, the composite price averages roughly € 3 260 per m², with a standard deviation of about € 1 590 and a range that spans from under € 900 to over € 13 000 per m². This wide dispersion is the first indication of substantial regional heterogeneity in Austrian housing markets. The house and apartment series differ markedly in level, with average house prices around € 4 200 per m² exceeding average apartment prices of about € 2 870 per m². Both series are right-skewed, so that a small number of high-price districts sit well above the median. The evidence presented is descriptive and is not intended to support causal claims, since the differences reported can reflect urbanity, dwelling composition, income, and other district characteristics rather than the transmission of monetary policy. The purpose is to establish that the data contain economically meaningful regional heterogeneity in price levels, price dynamics, and supply conditions, and to surface the features that shape the specification and interpretation of the local projections analysis.

The price dispersion is closely related to supply conditions. Grouping districts by the supply-constraint classification (see Section 4.3), the mean composite price in inelastic districts is roughly € 4 410 per m² against about € 2 480 in elastic districts, so that the more supply-constrained districts operate in a distinctly higher price regime. As

emphasized above, this level difference is not evidence of differential monetary transmission, since inelastic districts are also more urban and more apartment-heavy. It does, however, confirm that the supply classification separates districts that differ systematically in their housing markets, which is a precondition for the interaction analysis that follows.

More important for the empirical strategy is the time-series behavior of price growth. Quarterly composite price growth averages about 1.3% over the sample, with elastic districts growing marginally faster on average than inelastic districts. Pooling the within-district changes, the autocorrelation of quarterly composite growth is positive and persistent, at about 0.23 at the first lag and remaining close to 0.20 through the fourth lag. Positive autocorrelation means that a quarter of above-average price growth tends to be followed by further above-average growth, so that shocks to prices are not reversed immediately but propagate over several quarters. Price momentum is a well-documented feature of housing markets (Case & Shiller, 1988; Guren, 2018), and reflects the slow diffusion of information, search frictions, and the staggered nature of transactions in illiquid property markets.

The presence of price momentum has a direct methodological implication. If lagged price changes predict current growth and are also correlated with the timing of monetary policy surprises, omitting them would confound the estimated response with the ongoing momentum of the local market. Following Aastveit and Anundsen (2018), who include four lags of house-price changes in their panel local projections precisely to absorb this momentum, the control vector therefore includes four lagged quarterly changes of the composite price. The observed lag profile, which stays economically meaningful up to the fourth lag, supports the choice of four lags rather than a shorter window. The same persistence, combined with the overlapping cumulative outcomes of the local projections, also suggests the serial-correlation concern that governs the treatment of standard errors discussed in Section 5.

4.2. Monetary Policy Shocks

The monetary policy shock is constructed from the Euro Area Monetary Policy Event-Study Database of Altavilla et al. (2019), which records high-frequency changes in a range of asset prices within tight windows around ECB policy announcements. The

chosen instrument is the surprise in the twenty-year overnight index swap rate (OIS 20Y). This maturity is selected because residential real estate prices depend on expected long-term financing conditions rather than on the current policy rate. Housing is a long-lived asset whose price equals the present value of its future returns, discounted at a long-term rate, so that changes in long-term interest rates revalue property more strongly than changes in short rates (He & La Cava, 2020), a channel that operates through housing wealth in the euro area (Di Casola & Grothe, 2026). Consistent with this, the transmission of ECB policy, and of its asset purchase programmes in particular, works to a large extent through the long end of the yield curve (Eser et al., 2019), so that surprises at longer maturities are the component of monetary policy most relevant for housing valuations. The twenty-year and thirty-year German government bond yield surprises are retained only as alternative instruments for robustness.

The event-level surprises are aggregated to the quarterly frequency of the panel by summing all surprises within a quarter into a net quarterly shock. Alongside the net measure, the sum of positive surprises, the sum of negative surprises, and the sum of absolute surprises are recorded for each quarter, which supports the signed decomposition used to examine asymmetric responses. The sign convention follows the interpretation of the underlying instrument, a negative OIS 20Y surprise denotes a downward revision of long-term rate expectations and therefore a monetary easing surprise. This convention matters for the interpretation of the estimated coefficients and is carried through consistently in Section 5. Over the sample, the mean net quarterly surprise is negative, at roughly -0.59, consistent with the prevailing monetary easing stance during much of the period, and the largest absolute net surprises occur in Q3 2019 and around the onset of the tightening cycle in 2022 and 2023.

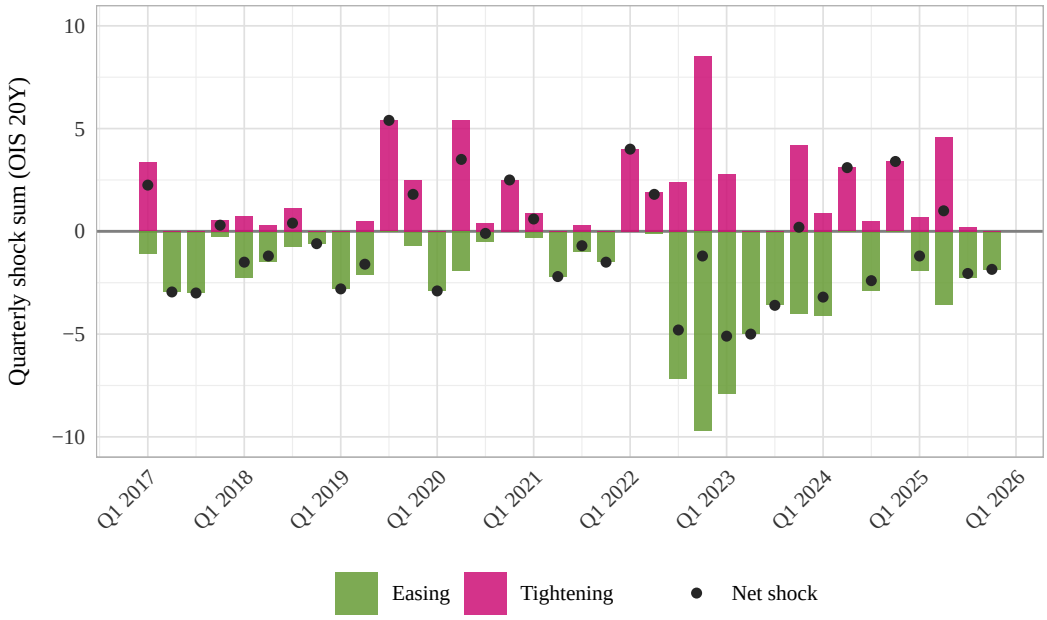


Figure 2: Quarterly monetary policy shock components
Easing surprises denoted as negative, tightening surprises denoted as positive; points = net quarterly sum

A preliminary diagnostic examines how far the quarterly net sum conceals offsetting surprises within a quarter. The cancellation ratio, defined as the net shock (net sum of shocks in a quarter) divided by the absolute sum of surprises in a quarter, has a mean of about 0.69, which indicates that easing and tightening surprises frequently occur within the same quarter and partly offset in the net measure. The ratio is a diagnostic rather than an additional regressor, and the extent of within-quarter cancellation motivates the asymmetric specification that separates the easing and tightening components of the shock (see Section 5.7).

Both easing and tightening quarters are well represented, with an easing component present in 31 of the 36 quarters and a tightening component in 26, and both signs occurring within the same quarter in 21 quarters. This overlap is the source of the within-quarter cancellation documented above, and the presence of both directions of surprise is what makes the sample suitable for examining whether price responses are asymmetric across monetary policy phases. Because the shock is common to all districts within a quarter, its aggregate effect is absorbed by the time fixed effects, and the descriptive variation reported here bears on the identifying interaction only through its

differential incidence across the supply groups.

4.3. Supply-Constraint Measure

Regional differences in housing supply are captured through the district building land reserve recorded in the ÖROK monitoring of building land reserves (ÖROK Monitoring of Building Land Reserves, 2025), which measures the share of vacant developable land in total designated building land. A lower reserve indicates less room for additional construction and therefore a more supply-inelastic district. To predate the estimation window, the 2014 level of the reserve is used throughout, so that the classification cannot respond to price developments during the sample.

Districts are classified as supply-inelastic using the twenty-fifth percentile of the reserve distribution outside Vienna, a threshold of around 22%, which is then applied to all districts. This procedure classifies 47 districts as inelastic and 69 as elastic. Of the inelastic districts, 24 lie outside Vienna and 23 are Viennese municipal districts. The measure is a proxy for supply conditions rather than a directly estimated supply elasticity, and it is strongly structured along geographic and compositional lines. In particular, the reserve measure correlates at about 0.91 with the fixed house weight, so that inelastic districts tend also to be apartment-heavy. The interpretation of this entanglement is developed alongside the correlation evidence in Section 4.4. District fixed effects absorb time-invariant level differences of this kind, but the entanglement of supply constraints with geography and property composition remains a concern for interpretation, which is why the analysis reports property-specific outcomes and a continuous reserve measure as alternative specifications, and re-estimates the model on the sample excluding Vienna as a robustness check (see Section 5.7).

The building land reserve varies considerably across districts, with a median of about 24% and values ranging from close to zero to above 45%. The supply classification is strongly clustered by federal state. All Viennese districts and all Salzburg districts fall into the inelastic group, while Burgenland contains no inelastic districts, and the median reserve in Vienna, at under 5%, is far below that of the predominantly rural states. This geographic concentration is economically intuitive, since the most supply-constrained districts are the dense urban and Alpine areas where developable land is genuinely scarce, but it also implies that the supply measure is entangled with region

and with property composition.

This entanglement carries two implications that are taken up in the empirical analysis. First, district fixed effects remove time-invariant differences in price levels and composition, but they do not by themselves rule out that districts differing in these correlated characteristics also respond differently to a common shock for reasons other than supply elasticity. Second, because the interaction with the shock is by construction non-zero only for inelastic districts, and because all Viennese districts are inelastic, the identifying variation is concentrated in a structurally distinctive set of markets. These considerations motivate the property-specific outcomes, the Vienna-excluded specification, and the continuous reserve measure that are reported as robustness checks, and they caution against reading the baseline interaction as a clean measure of supply elasticity alone.

4.4. Control Variables

The control set comprises district-level population, the unemployment rate, and net income per capita, each aligned to the quarterly frequency of the panel. Population is available quarterly from the central residence register and enters directly (Statistik Austria, 2026b). Unemployment is available annually at the district level and is carried across the four quarters of each year (Statistik Austria, 2026a). Net income per capita is constructed by dividing annual district net income totals (Statistik Austria, 2026d) by the corresponding population, so that within-year variation in income per capita partly reflects population movements. This construction creates a mechanical correlation between income growth and population growth. Since these series are annual in origin, the quarterly panel repeats the annual figures within each year, and this is reflected in the way the control lags behave.

Several series require construction at the tail of the sample, where source data are not yet available. Annual net income is observed through 2022 and is extrapolated for 2023 to 2025 using the nominal Tariflohnindex (OeNB - Österreichische Nationalbank, 2026), so that all districts share a common wage-index growth path over the extrapolated years. District unemployment is observed through 2024 and carried forward to 2025, and population is carried forward from the first quarter of 2025 to the remaining quarters of that year. These constructed values are complete observations, but they do not represent newly observed district information. Their share is largest at the end of the sample and

declines at longer horizons as later quarters drop out of the estimation window. To ensure that these choices do not drive the findings, the analysis reports a sensitivity sample that excludes the constructed tail.



Figure 3: Pairwise correlations of key variables

Figure 3 reports the pairwise correlations among the outcome, the controls, the shock and its signed components, the supply measures, and the shock-by-inelastic interaction. The matrix serves two purposes, it documents that the shock behaves as an external regressor with respect to district characteristics, and it surfaces one strong association that shapes the interpretation of the results.

The most reassuring result for identification is the relation of the shock to the district-

level variables. The shock is essentially uncorrelated with the two time-invariant supply measures, with a coefficient at 0.00 with both the building land reserve and the fixed house weight. Further, it correlates only weakly with the district controls, at −0.08 with income growth, −0.02 with population growth, and −0.22 with the change in unemployment. The near-zero correlation with the time-invariant measures is mechanical rather than coincidental, since a common national surprise carries no cross-sectional variation that could line up with fixed district traits, but it nonetheless confirms that the identifying variation in the shock does not proxy for the supply classification. The modest negative correlation with the unemployment change is consistent with monetary easing coinciding with weaker labor market conditions, and it motivates the inclusion of the controls rather than raising a collinearity concern.

The shock also relates to the outcome in the expected direction (see Section 5.4). The net shock correlates at 0.19 with quarterly composite price growth, where the signed decomposition is more informative, with the easing component correlating at 0.32 and the tightening component at −0.12. Under the convention that easing surprises enter negatively, these signs indicate that price growth is higher in quarters with larger easing surprises, which is the unconditional counterpart of the response the local projections estimate conditionally. Within the shock block, the net measure correlates at 0.66 with the easing component and at 0.55 with the tightening component, while the two signed components correlate at −0.26 with each other, reflecting the within-quarter cancellation documented in Section 4.2. The interaction term correlates at 0.63 with the net shock, as expected since it is the shock scaled by the inelastic indicator, and only weakly, at around 0.13 to 0.14, with the two supply measures.

Among the controls, the strongest association is the −0.49 correlation between income growth and population growth. This is largely mechanical, as income per capita is constructed by dividing an annual district income total by quarterly population, so that movements in the denominator enter both series with opposite sign, as noted in Section 4.4. The remaining control correlations are small, with composite price growth correlating at −0.20 with the change in unemployment and at 0.15 with income growth, which is consistent with local demand conditions without indicating redundancy among the regressors.

The one association that requires closer attention is the 0.91 correlation between the building land reserve and the fixed house weight. Because inelastic districts are pre-

dominantly urban and apartment-heavy, the supply classification and the composite property mix carry closely related cross-sectional information, so that the inelastic indicator partly captures the difference between apartment markets and house markets rather than land scarcity alone. This entanglement is the central measurement concern of the analysis.

By construction, apartments carry more weight in the housing stock of urban and therefore inelastic districts. Simultaneously, apartment-heavy markets are more strongly shaped by investor demand than owner-occupied house markets. This matters because the two segments do not respond to monetary policy in the same way. Biljanovska et al. (2025) document that investor demand, measured through non-owner-occupied purchases, contracts more sharply after monetary tightening, a pattern they attribute to investors extending recent price trends into their expectations of future returns, which makes their demand particularly sensitive to financing conditions. Investors evaluate housing as one asset among others and discount expected future returns at market interest rates, so that a shift in the policy stance revalues investor-held property more forcefully than owner-occupied housing, where the decision to buy or sell is tied to housing needs rather than to the rate of return (He & La Cava, 2020). The behavioral evidence points in the same direction, as Genesove and Mayer (2001) find that owner-occupants adjust their asking prices far more reluctantly than investors when facing nominal losses, so that owner-occupied house markets are the stickier of the two segments.

This literature therefore indicates that the apartment-heavy composition of inelastic districts is not merely a statistical nuisance but a channel that plausibly reinforces the supply mechanism, since the districts classified as inelastic are also those in which the more monetary-sensitive investor segment is most prominent. The two channels nonetheless cannot be fully separated within the current design, as the inelastic indicator jointly reflects land scarcity and property composition. The analysis therefore does not claim that the interaction isolates a pure supply-elasticity effect, and it addresses the concern directly through the property-specific outcomes and the continuous reserve measure reported as alternative specifications, together with the Vienna-excluded robustness check, which are discussed in detail in Section 5.7.

4.5. Panel Construction and Preliminary Diagnostics

The estimation panel is assembled by merging the price series, the composite weights, the supply-constraint classification, the quarterly shock, and the three control variables on the district and quarter keys. The price series and controls vary by district and quarter, the shock is common across districts within a quarter, and the supply classification and composite weights are time-invariant district characteristics. Districts are identified by a consistent numeric code across all sources, and the Viennese municipal districts are aligned to the three-digit district scheme used throughout the panel.

A set of preliminary diagnostics confirms that the merge preserves the intended structure of the data. All component inputs join without duplicate or unmatched keys, every district appears in every quarter, and the price series contain no non-positive or non-finite values, so the merged panel is balanced and complete at 4 176 district-quarter observations (covering 116 districts over 36 quarters). Summary statistics for the outcome, the shock, the supply measure, and the controls, together with the integrity checks underlying these statements, are reported in Appendix 1. The descriptive evidence discussed above draws on this panel, and the diagnostics reported here establish that the regularities documented in the preceding subsections reflect the data.

The dependent variable and the lagged controls introduce structural missing values by construction rather than through gaps in the source data. The cumulative outcome (as described in Section 5.3) at horizon h is measured relative to the quarter preceding the shock, and four lagged quarterly changes enter as controls, so the usable sample declines from 3 596 observations at the shortest horizon to 2 204 at the longest. All 116 districts remain represented at every horizon, but each additional horizon removes the most recent quarter, so that comparisons of coefficients across horizons also compare progressively earlier samples. This feature is inherent to the local projection design and is disclosed alongside the estimates.

5. Empirical Strategy

The following chapter describes how the district-quarter panel is used to estimate how the dynamic response of real estate prices to ECB monetary policy shocks differs between supply-elastic and supply-inelastic districts. The approach is a panel local projections

analysis, which traces the price-response over a three-year horizon and identifies the differential effect through the interaction of the shock with the supply-constraint classification. The following subsections introduce the local projection method, specify the baseline model and its identifying assumptions, explain the interpretation of the coefficient of interest, describe the outcome transformation and the fixed-effects structure, and discuss the treatment of inference. Further, the concluding subsection in this chapter discusses alternative model specifications and robustness checks.

5.1. Panel Local Projections

The question motivating this thesis is inherently dynamic, since the interest lies not only in whether prices react to a monetary policy shock but in how that reaction unfolds over the quarters that follow. A common way to trace such a reaction is vector autoregression, which summarizes the joint dynamics of a set of variables in a single system and then projects that system forward to obtain the response at each horizon. The drawback is that the response at a distant horizon is built from the same estimated system used at the nearest one, so that any error in how the system is specified accumulates as the horizon grows (Jordà, 2005). For a study concerned with responses up to three years ahead, this accumulation is a concern.

Local projections offer a more direct route and are the method adopted here. Rather than relying on a single system, a separate regression is run for each horizon, relating the price outcome several quarters ahead to the shock and to controls measured at the time of the shock, and the response is obtained from the coefficients on the shock across this sequence of regressions (Jordà, 2005). The appeal of this method for the setting of this thesis is threefold. It is estimated by ordinary least squares (OLS) and thus extends without difficulty to a panel with district and time fixed effects. Because each horizon is estimated separately, a misspecification of the dynamics at one horizon does not contaminate the others. And it accommodates the interaction between the shock and a district characteristic that lies at the center of this analysis, which would be far less straightforward to impose within a full multivariate system. One drawback is that the outcomes used across horizons overlap in time, which leaves the residuals serially correlated and requires care in the treatment of standard errors, as further discussed in Section 5.6.

These properties carry over naturally to a panel, for which Jordà (2023) describes local projections as a way of connecting the impulse-response tradition of macroeconomics with the treatment-effect reasoning of applied microeconomics, with the shock playing the role of an intervention whose effect is followed through time. In this thesis the panel does more than accommodate the method. Since the shock is the same for every district in a given quarter, its overall effect on Austrian real estate prices cannot be separated from anything else happening nationally in that quarter, and it is not the object of interest. What can be identified is how districts differ in their reaction according to their supply conditions, which follows from interacting the common shock with the district-level classification. The construction of that interaction, and the fixed-effects structure that isolates the differential it captures, are developed in the following subsections.

5.2. Baseline Model and Identification

The baseline specification follows the panel local projection design of Aastveit and Anundsen (2018), adapted to the Austrian district-level setting. For each horizon $h = 0, 1, \ldots, 12$, a separate regression is estimated of the form:

$$y_{i,t+h} - y_{i,t-1} = \alpha_{i,h} + \tau_{t,h} + \beta_h \left(\text{Shock}_t \times \text{Inelastic}_i\right) + \theta_h' X_{i,t} + \varepsilon_{i,t+h}, \tag{5.1}$$

where i indexes districts and t quarters. The dependent variable $y_{i,t+h} - y_{i,t-1}$ is the cumulative change in the log composite price between the quarter preceding the shock and horizon h, so that the sequence of coefficients traces the response of prices over a maximum period of three years. The outcome construction is described in detail in Section 5.3. The horizon runs from the quarter of the shock, $h = 0$, to twelve quarters ahead, which spans the three-year window over which the price response is of interest.

The regressor of interest is the interaction $\text{Shock}_t \times \text{Inelastic}_i$, formed from the quarterly net OIS 20Y surprise and the time-invariant supply classification. Here $\text{Inelastic}_i \in \{0, 1\}$ is an indicator equal to one for districts classified as supply-inelastic and zero otherwise, constructed as described in Section 4.3. Its coefficient β_h is the object the analysis seeks to estimate, and it measures how the price response at horizon h differs between inelastic and elastic districts. The term $\alpha_{i,h}$ denotes district fixed effects and $\tau_{t,h}$ quarter fixed effects, both estimated separately at each horizon; their role in isolating the differential

response is discussed in Section 5.5. The vector $X_{i,t}$ collects the control variables, entered as four quarterly lags each of the change in the log composite price, the change in log income per capita, population growth, and the change in the unemployment rate, following the lag structure motivated in Section 4.1.

Two features of this specification deserve emphasis. First, the shock enters only through its interaction with the supply classification, not on its own. Because the surprise is common to all districts within a quarter, its level is collinear with the quarter fixed effects and cannot be separately estimated, so the specification is designed to recover the differential response rather than the aggregate one. Second, the coefficient of interest is allowed to vary freely across horizons, since each β_h is estimated in its own regression, which is what permits the response to build, peak, and fade over the three-year window without imposing a particular dynamic shape. The interpretation of β_h, including the role of the sign convention on the shock, is developed in Section 5.4, and the assumptions required for it to carry a causal reading are discussed in Section 5.6.

5.3. Outcome Transformation

The dependent variable is a cumulative log-price change, which is the standard outcome in the local projection literature (Jordà, 2005, 2023) and the one used by Aastveit and Anundsen (2018). Let $p_{i,t}^{\text{comp}}$ denote the composite price of district i in quarter t, constructed as described in Section 4.1, and let $y_{i,t} = \ln p_{i,t}^{\text{comp}}$ be its natural logarithm. The outcome at horizon h is the change in this log price between the quarter preceding the shock and h quarters after it,

$$y_{i,t+h} - y_{i,t-1} = \ln p_{i,t+h}^{\text{comp}} - \ln p_{i,t-1}^{\text{comp}}, \tag{5.2}$$

so that a separate outcome is formed for each horizon $h = 0, 1, \ldots, 12$. Because it is a difference in logarithms, the outcome measures the cumulative growth of the composite price from the pre-shock quarter to horizon h, and for the moderate changes observed over the sample it is closely approximated by the corresponding percentage change. Expressing the response as a proportional change also places every district on a common percentage scale, so that a euro movement counts relative to the local price level rather than in absolute terms. This matters in a panel where the composite price ranges

from under € 900 to over € 13 000 per m², since it prevents the response from being mechanically larger in the most expensive districts. The district fixed effects remove differences in the price *level*, whereas the log transformation governs the scale on which the change itself is measured.

Two aspects of this construction shape the interpretation of the estimates. First, the outcome is measured relative to quarter $t-1$ rather than quarter t. Anchoring the base in the quarter before the shock ensures that the horizon-zero outcome, $y_{i,t} - y_{i,t-1}$, already captures the immediate within-quarter movement in prices, so that the response is measured from a point that is not itself contaminated by the shock. Second, the outcome is cumulative rather than a period-by-period change. Each β_h therefore describes the differential response of the price *level* by horizon h, measured from its pre-shock value, rather than the differential growth in that single quarter. The sequence $\{\beta_h\}_{h=0}^{12}$ consequently traces the full path of the accumulated price response over the three-year window, which is the object of the temporal sub-question introduced in Section 1.

Forming the outcome from leads of the log price mechanically removes the final quarters of each district from the longer-horizon regressions, since the price twelve quarters ahead is not observed for the most recent quarters of the sample. The resulting decline in the estimation sample with the horizon, and its implication that longer-horizon coefficients rest on progressively earlier quarters, is documented in Section 4.5 and is revisited when the results are interpreted.

5.4. Interpretation of the Coefficient

The coefficient β_h is the central estimate of the analysis, and its interpretation follows directly from the structure of the model. Since the shock enters only through its interaction with the inelastic indicator, β_h measures the difference in the cumulative price response at horizon h between a supply-inelastic and a supply-elastic district exposed to the same monetary policy surprise. In the language of dynamic treatment effects, to which Jordà (2023) connects the local projection, β_h is the additional effect of the shock on inelastic districts relative to the elastic benchmark, traced out over the three-year window. It is not the response of any district in isolation, and it is not the average response across all districts, both of which are bound up with the common time effect and cannot be recovered from this specification.

The sign of β_h must be read together with the sign convention on the shock, and the two combine in a way that is easily misread. A negative OIS 20Y surprise denotes monetary easing (see Section 4.2), so the shock variable takes negative values in easing quarters. The differential effect on the cumulative price response of an inelastic district is the product $\beta_h \times \text{Shock}_t$. If, as the mechanism in this thesis suggests, easing raises prices more strongly where supply is constrained, then this product must be positive in an easing quarter, which requires β_h to be negative. A negative estimate of β_h therefore supports the hypothesis of a stronger response in inelastic districts. To avoid this ambiguity, the results are reported and discussed in terms of the implied response to a monetary easing surprise of a given size, rather than in terms of the raw coefficient alone.

This differential reading is the direct empirical counterpart of the research question, and it mirrors the approach of Aastveit and Anundsen (2018), who interpret the interaction between a monetary policy shock and a measure of supply elasticity as the extent to which tighter supply amplifies the price response. The magnitude of β_h is expressed in the units of the outcome, so that it represents the differential cumulative change in the log composite price, in log points, by horizon h, in response to a one-unit surprise. The full sequence $\{\beta_h\}$ describes how this differential builds and evolves over the twelve quarters, which speaks to the temporal sub-question, while its sign and persistence speak to whether supply constraints amplify or merely delay the transmission of monetary policy to regional prices. Whether β_h can be given a causal rather than a purely associational reading depends on the identifying assumptions discussed in Section 5.6.

5.5. Fixed Effects

The specification includes both district and quarter fixed effects, estimated separately at each horizon, and each serves a distinct purpose. District fixed effects $\alpha_{i,h}$ absorb all time-invariant characteristics of a district, including its average price level, its position in the supply-constraint classification, and its fixed property composition. This matters given the concern raised in Section 4.4 that the inelastic indicator is closely tied to an apartment-heavy housing stock. By removing the cross-sectional level differences between districts, the district effects ensure that β_h is not driven by the fact that inelastic districts are more expensive or more urban on average, but by how their prices move

differently over time.

Quarter fixed effects $\tau_{t,h}$ are what allow the model to be read as a differential response. They absorb everything common to all districts within a given quarter, including the level of the monetary policy shock itself, the national housing cycle, euro-area financial conditions, and any other aggregate development. Because the shock does not vary across districts within a quarter, its level is perfectly collinear with the quarter fixed effects and drops out of the regression, as noted in Section 5.2. The only variation left to identify β_h is therefore cross-sectional and within-quarter, namely the difference in how inelastic and elastic districts respond to the same aggregate surprise. This is a deliberate feature rather than a limitation, since it removes at a stroke the large class of confounders that operate at the national level and would otherwise contaminate an estimate of the aggregate response.

This two-way structure follows Aastveit and Anundsen (2018), who combine regional fixed effects with the interaction between the shock and supply elasticity, and whose extended specification adds region-by-time effects to absorb common regional developments. In the present setting the quarter fixed effects already subsume all national variation, and the panel is not large enough along the cross-sectional dimension to add Bundesland-by-quarter fixed effects, the Austrian analogue of these region-by-time effects, without exhausting the identifying variation. The fixed effects carry the horizon subscript h because a separate regression is estimated at each horizon, so that the absorbed level differences and common shocks are allowed to matter differently at different points of the response path. In practice each horizon regression is estimated by ordinary least squares (OLS) with the two sets of fixed effects absorbed through the within transformation, which removes the district and quarter means without estimating a separate coefficient for each unit. What this structure cannot absorb are district-specific developments that are correlated with the supply classification and that happen to coincide with the timing of the shocks, and the plausibility of ruling these out is taken up in Section 5.6.

5.6. Inference and Identification Assumptions

Valid inference is the most delicate part of the design, because the residuals of a panel local projection are dependent along two dimensions at once. To make this concrete,

the baseline specification of Equation 5.1 was fit at selected horizons as a preliminary diagnostic, ahead of the full estimation, and the dependence in its residuals is reported in Appendix 3. The first dimension is temporal. Because the outcome at each horizon accumulates the price change from the same base quarter, the regressions at adjacent horizons share overlapping windows of data, which leaves the residuals serially correlated within districts. The within-district first-order autocorrelation of the residuals, that is, the correlation between a district's residual in one quarter and its residual in the next, rises from near zero at horizon zero to about 0.85 at the longest horizon. It grows with the horizon precisely because the longer-horizon outcomes accumulate over increasingly overlapping windows, so that the errors of neighboring horizons share progressively more of the same price movements. The second dimension is cross-sectional. Districts are exposed to the same national shock and the same aggregate housing cycle, and neighboring markets are linked through common regional developments, so the residuals are correlated across districts within a given quarter, with their mean absolute pairwise correlation reaching roughly 0.43 at the longest horizon. Ordinary standard errors, which assume independence, would understate the true sampling uncertainty on both counts and cannot be used.

Two families of correction address this dependence, and the analysis reports both rather than committing to one. The first is the estimator of Driscoll and Kraay (1998), which extends the nonparametric time-series correction of Newey and West (1987) to a panel and yields standard errors that remain valid under broad patterns of dependence both across districts and over time. Its asymptotic justification relies on a growing time dimension, which is a genuine limitation here, since the panel contains only 36 quarters. The second is two-way clustering by district and quarter in the manner of Cameron et al. (2011), which allows for arbitrary within-district and within-quarter correlation. Its asymptotics require the number of clusters in both dimensions to grow, and with only 36 quarterly clusters the time dimension is again the binding constraint, so the clustered errors are likely to be somewhat understated. A third, complementary route specific to local projections is lag augmentation, which Montiel Olea and Plagborg-Møller (2021) show renders ordinary heteroskedasticity-robust standard errors valid by absorbing the serial correlation induced by the projection structure, thereby removing the need for an explicit autocorrelation correction. Because none of these approaches is fully satisfactory at the sample size available, the standard errors are reported side by side, and inference

is treated as robust only where the conclusions do not depend on which correction is used. Statistical significance under a single estimator is not taken as sufficient grounds for a finding.

The corrections just discussed concern the precision of the estimates, that is, whether the standard errors around β_h are trustworthy. Reading β_h as a causal differential rather than a mere correlation is a separate matter, and it rests on three identifying assumptions that the fixed effects and the standard errors cannot by themselves secure.

(A1) Shock exogeneity. The monetary policy surprise must be exogenous with respect to Austrian district housing conditions. This is the rationale for the high-frequency identification of Altavilla et al. (2019), which isolates the component of policy that is unanticipated by measuring asset-price movements in a tight window around ECB announcements, so that the surprise is plausibly uncorrelated with developments already underway in regional housing markets. The main threat to this assumption is the central-bank information effect, whereby announcements also convey the central bank's assessment of the economic outlook (Jarociński & Karadi, 2020). Since the shock enters only in interaction with a district characteristic, however, any such aggregate information content is absorbed by the quarter fixed effects and does not by itself bias the differential.

(A2) No confounding district characteristic. No omitted district characteristic may be both correlated with the supply classification and differentially responsive to the common shock. This is precisely the class of confounder that the fixed effects cannot absorb, as noted in Section 5.5, and it is the reason the entanglement between supply inelasticity and property composition documented in Section 4.4 matters for interpretation rather than only for description. The analysis does not claim to rule this out by assumption, but confronts it through the alternative specifications and robustness checks as described in Section 5.7, which re-estimate the model on property-specific outcomes, on the sample excluding Vienna, and with a continuous reserve measure in place of the binary indicator.

(A3) Validity of the supply proxy. The building land reserve must be a valid proxy for the elasticity of local housing supply. As discussed in Section 4.3, the reserve is a predetermined measure of the scope for additional construction rather than a directly estimated elasticity. Taken together, assumptions (A1) to (A3) define the conditions

under which β_h can be read as the causal differential response of supply-inelastic districts, and the discussion of results returns to their plausibility where relevant.

5.7. Alternative Specifications and Robustness

The analysis reports a set of departures from the baseline specification that fall into two groups, distinguished by their purpose. Alternative specifications change the model of the effect itself and are reported alongside the baseline, because they characterize the response rather than merely testing its durability. Robustness checks instead vary an estimation or sample choice while leaving the effect of interest unchanged, in order to show that the baseline finding is not an artifact of a particular decision. The distinction follows the practice of Aastveit and Anundsen (2018), and each of these departures is motivated by a specific feature of the data documented in Section 4.

Three alternative specifications are considered. **(AS1) Property-specific outcomes.** The model is re-estimated with the house and apartment series as separate outcomes in place of the composite, which tests whether the composite result is an artifact of the fixed property-mix weighting and speaks to the entanglement between supply inelasticity and property composition discussed in Section 4.4. **(AS2) Asymmetric shock decomposition.** The net shock is decomposed into its easing and tightening components, which is required to address the sub-question on whether the transmission of monetary policy is asymmetric across the phases of the cycle. For each horizon h, this specification replaces the single interaction in Equation 5.1 with two,

$$\begin{aligned} y_{i,t+h} - y_{i,t-1} = \alpha_{i,h} + \tau_{t,h} + \beta_h^{\text{ease}} \left(\text{Shock}_t^{\text{ease}} \times \text{Inelastic}_i\right) \\ + \beta_h^{\text{tight}} \left(\text{Shock}_t^{\text{tight}} \times \text{Inelastic}_i\right) + \theta_h' X_{i,t} + \varepsilon_{i,t+h}, \end{aligned} \tag{5.3}$$

where $\text{Shock}_t^{\text{ease}}$ and $\text{Shock}_t^{\text{tight}}$ are the within-quarter sums of the negative and positive OIS 20Y surprises respectively, so that β_h^{ease} and β_h^{tight} measure the differential response of inelastic districts to easing and tightening surprises separately. **(AS3) Continuous supply measure.** The binary inelastic indicator is replaced with the continuous building land reserve, which removes the dependence of the result on the classification threshold and uses the full cross-sectional variation in supply conditions.

Five robustness checks probe the durability of the baseline. **(R1) Vienna excluded.** The model is re-estimated on the sample excluding the Viennese districts, since all of Vienna falls into the inelastic group and its apartment-dominated markets are structurally distinctive, so this check confirms that the differential is not driven by Vienna alone. **(R2) Pre-tail sample.** The model is re-estimated on the sample that excludes the constructed tail of the control series, where income is extrapolated and unemployment and population are carried forward, as documented in Section 4.4, to confirm that the imputed observations do not drive the findings. **(R3) Alternative shock instruments.** The twenty-year and thirty-year German government bond yield surprises are used in place of the OIS 20Y instrument, which checks that the result does not hinge on the particular measure of the shock. **(R4) Alternative classification threshold.** The building-land threshold that separates elastic from inelastic districts is varied, which complements the continuous-measure specification in guarding against an arbitrary cut-off. **(R5) Standard-error comparison.** The standard errors are compared across the estimators described in Section 5.6, so that a finding is treated as robust only where it does not depend on the correction used. Checks that feature in the regional local projection literature but that the present data cannot support, such as the balanced-versus-unbalanced comparison of Aastveit and Anundsen (2018), are not applicable here, since the panel is balanced by construction, and district-by-district estimation is precluded by the short time dimension.

6. Empirical Results

6.1. Baseline Results

6.2. Alternative Responses

6.3. Robustness Checks

7. Discussion

7.1. Interpretation of Findings

7.2. Relation to Literature

7.3. Economic Relevance

7.4. Policy Implications

7.5. Limitations

8. Conclusion

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A. Appendix

Variable	N	Mean	SD	Min	Median	Max
Outcome						
Composite price (€/m^2)	4 176	3 261	1 593	868	2 876	13 500
House price (€/m^2)	4 176	4 203	2 699	849	3 200	16 212
Apartment price (€/m^2)	4 176	2 865	1 539	742	2 471	13 515
Composite price growth (log change)	4 060	0.0135	0.0238	-0.1080	0.0118	0.1570
Monetary policy shock						
Net shock (OIS 20Y)	36	-0.59	2.66	-5.10	-1.20	5.40
Easing component	36	-2.29	2.30	-9.70	-1.90	0.00
Tightening component	36	1.70	2.06	0.00	0.72	8.50
Supply measure						
Building land reserve (%)	116	21.2	10.2	0.1	23.8	45.5
House weight	116	0.500	0.273	0.003	0.580	0.874
Controls						
Population	4 176	77 446	47 622	1 900	65 370	305 314
Unemployment rate (%)	4 176	6.0	3.3	1.6	5.0	17.9
Net income per capita (€)	4 176	22 680	4 797	5 155	21 680	51 092

Note. N differs by variable block. Prices and controls are observed at the district-quarter level. The monetary policy shock varies only over the 36 quarters. The supply measure and the fixed dwelling-stock weights are time-invariant, they vary over the 116 districts. Composite price growth is a quarterly log change and is undefined in the first quarter of each district, which accounts for its lower count.

Table 1: Summary statistics of the analysis variables

Diagnostic	Result
Balanced panel (districts × quarters)	116 × 36 = 4 176
Unique district-quarter keys	4 176 (no duplicates)
Missing or unexpected keys	0
Districts observed in all 36 quarters	116
Source inputs joined without key errors	7 of 7
Non-positive or non-finite price cells	0
Weight-sum and composite identity errors	0

Table 2: Panel construction and integrity diagnostics

Horizon h	Residual AR(1)	Mean $\lvert\rho\rvert$	Max $\lvert\rho\rvert$
0	-0.01	0.19	0.81
4	0.78	0.30	0.98
8	0.86	0.41	0.99
12	0.85	0.43	0.99

Notes. Residuals are from the baseline two-way fixed-effects specification (Equation 5.1), fit at selected horizons h as a preliminary diagnostic ahead of the full estimation. Residual AR(1) is the first-order within-district autocorrelation of the residuals; Mean $\lvert\rho\rvert$ and Max $\lvert\rho\rvert$ are the mean and maximum absolute pairwise correlation of the residuals across districts within a quarter. The rising values over horizons reflect the serial and cross-sectional dependence discussed in Section 5.6.

Table 3: Residual dependence of the two-way fixed-effects specification